TÉCNICO
UNIVERSIDADE
DE LISBOA

0.6 · SHIP AND OCEAN SYSTEMS DESIGN

The Reference Ship

37,000 tdw Product/ Chemical Tanker

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The Reference Ship: Owner's Brief

Every in-class exercise uses this ship, a 37,600 t product and chemical tanker based on a real ship built in 2015. Your project uses your Baseline ship (0.2, slides 4 and 5).

The ship	
Type	Oil/chemical tanker, double hull, IMO ship type II; single screw; machinery and deckhouse aft
Deadweight	37,600 t at the summer draught
Cargo tanks	14 tanks, 39,800 m³ (100 %); 2 slop tanks, 2,700 m³
Service speed	14.5 kn at the design draught of 9.8 m, engine at its normal rating, 15 % sea margin
Main engine	One low-speed two-stroke diesel (MAN B&W ME-B type)
Fuel	VLSFO for the main engine, MGO for the generators
Class and flag	KR · Marshall Islands
Rules	SOLAS, MARPOL Annexes I, II and VI (EEDI for tankers), IBC Code, Load Lines (type A freeboard)
Crew	25, plus 6 extra berths

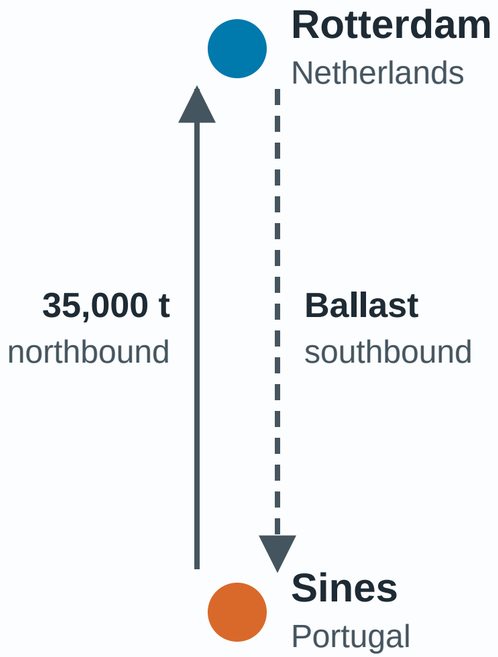
The trade	
Route	Sines – Rotterdam – Sines, 350 operating days a year
Cargo per round voyage	35,000 t of refined products northbound, ballast southbound
Cargo densities	0.70–1.25 t/m³; design 0.90 t/m³
Cargo handling	Loading about 24 h at Sines (shore pumps); discharge about 24 h at Rotterdam (ship's pumps)
Terminal limits	Draught 11.5 m at both; length and beam not limiting
Range	10,000 nm at service speed; bunkers at Rotterdam

Classed by KR: What the Notation Says

Hull	Meaning
+ KRS1	Hull built under KR survey; unrestricted service area
Oil/Chemical Tanker (Double Hull)	Ship type; double hull around the cargo tanks
ESP	Enhanced Survey Programme applies
CSR	IACS Common Structural Rules for oil tankers
FBC	Cargoes with flash point at or below 60 °C, controlled tank venting
Product / II 2G 1.025SG (IBC)	Product oil as main cargo; chemical ship type II; integral tanks; cargo density up to 1.025; IBC Code
Sea Trust (HCM)	KR hull condition monitoring programme
IWS	In-water survey: bottom surveyed afloat
PSPC	Ballast tank coatings to the IMO performance standard
LI	Loading instrument on board
CHA	See the KR Notation Guide

Environment and machinery	Meaning
ERS	Emergency response service
ENV	Environmental notation, covering:
IBWM · IAFS	Ballast water management · anti-fouling system
IOPP · ISPP	Oil · sewage pollution prevention
IGPP · IAPP	Garbage · air pollution prevention
IIHM · IEE	Hazardous materials inventory · energy efficiency
VEC-L	Vapour emission control system
+ KRM1	Machinery built under KR survey; unrestricted service area
UMA	Unattended machinery space
STCM	See the KR Notation Guide
IGS	Inert gas system for the cargo tanks

A Product Shuttle Between Two Terminals



	Sines	Rotterdam
Draught limit	11.5 m	11.5 m
Cargo handling	Load in about 24 h	Discharge in about 24 h
Pumps	Shore	Ship’s own
Each call	Load 35,000 t, sail laden	Discharge 35,000 t, sail in ballast

Round voyage Sines – Rotterdam – Sines, **350 operating days a year**, at **14.5 kn**. Laden northbound, in ballast southbound.

> Port data are course assumptions, set for the exercises.